Networking Layer (VPC Design)

Public Subnet 1:

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public Subnet A

The name can be up to 256 characters long.

Availability Zone Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az2 (us-east-1a)

IPv4 VPC CIDR block Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.1.0/24 256 IPs

Public Subnet 2:

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public Subnet B

The name can be up to 256 characters long.

Availability Zone Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az4 (us-east-1b)

IPv4 VPC CIDR block Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.2.0/24 256 IPs

Private Subnet 1:

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Private Subnet A

The name can be up to 256 characters long.

Availability Zone Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az2 (us-east-1a)

IPv4 VPC CIDR block Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.3.0/24 256 IPs

Private Subnet 2:

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Private Subnet B

The name can be up to 256 characters long.

Availability Zone Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az4 (us-east-1b)

IPv4 VPC CIDR block Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.4.0/24 256 IPs

Internet Gateway (IGW)

VPC > Internet gateways > Create internet gateway

Create internet gateway Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

Enterprise-IGW

Tags - *optional*

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - *optional*	
Name	Enterprise-IGW	Remove

Add new tag

You can add 49 more tags.

NAT Gateway

Create NAT gateway Info

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - *optional*

Create a tag with a key of 'Name' and a value that you specify.

Enterprise-NGW

The name can be up to 256 characters long.

Availability mode Info

Choose whether to deploy across all zones in the region or restrict to a single availability zone.

- (●) Regional - *new*
  Scales automatically across all regional AZs, simplifying management for multi AZ deployments.
- ( ) Zonal
  Provides granular control within a specific availability zone, adhering to subnet level settings.

VPC

Select a VPC in which to create the regional NAT gateway.

vpc-04039803f5bcd1e4d (Enterprise-VPC)

Connectivity type

Select a connectivity type for the NAT gateway.

- (●) Public
- ( ) Private

Method of Elastic IP (EIP) allocation Info

Choose how IP addresses are associated with NAT gateways.

- (●) Automatic
  AWS automatically manages EIPs and AZ coverage for NAT gateways. This ensures easy scaling-- adding AZs automatically allocates EIPs, simplifying management.
- ( ) Manual
  Manually assigns specific IP addresses for compliance or whitelisting. Note: Requires manual scaling to new AZs as workloads expand.

Route Tables

Route table rtb-06081eb177d76671b | Private-RT was created successfully.

Route tables (2/4) Info

Last updated less than a minute ago | Actions | Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...
–	rtb-0dc62f531b68e1372	–
Public-RT	rtb-0e8ef347f2b1150d5	2 subnets
–	rtb-0fcad194e49b3a5cb	–
Private-RT	rtb-06081eb177d76671b	–

Security groups:

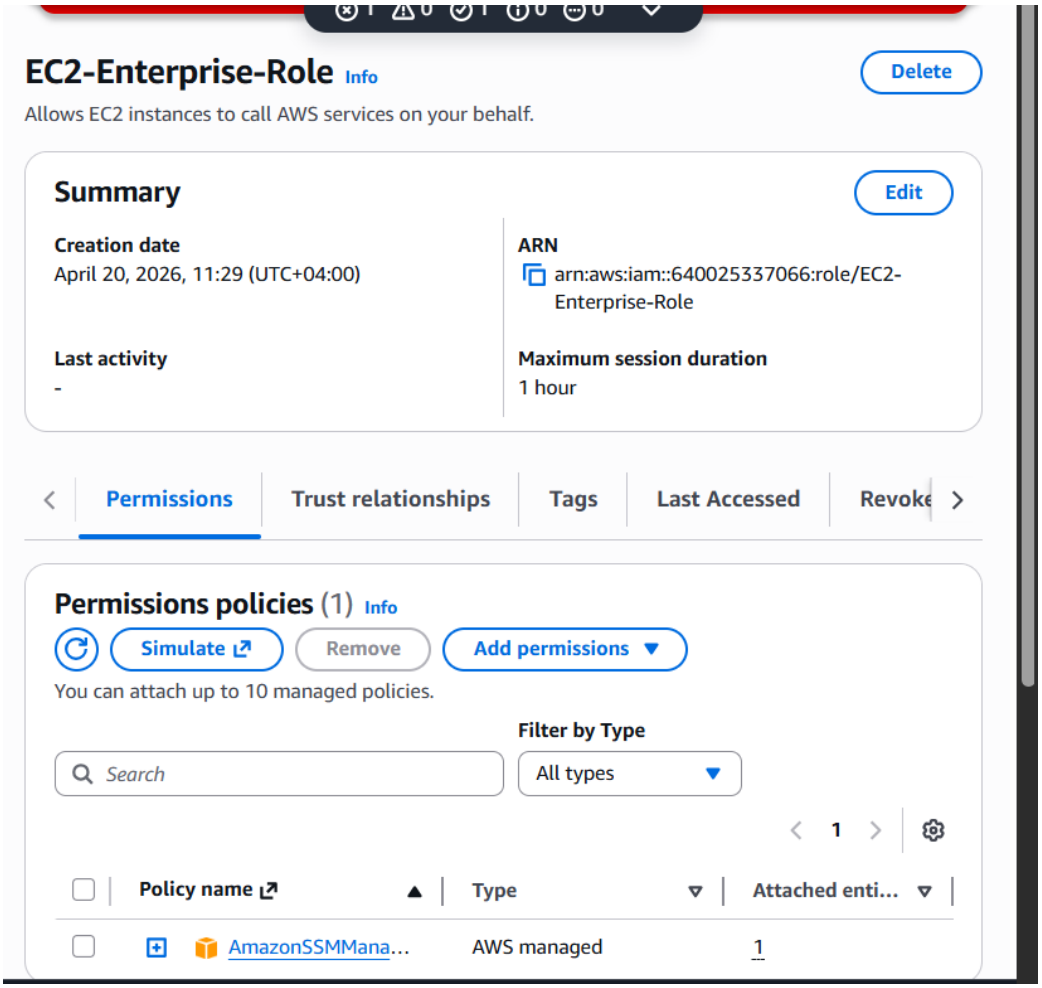
Security Groups (5) Info

Actions | Export security groups to CSV | Create security group

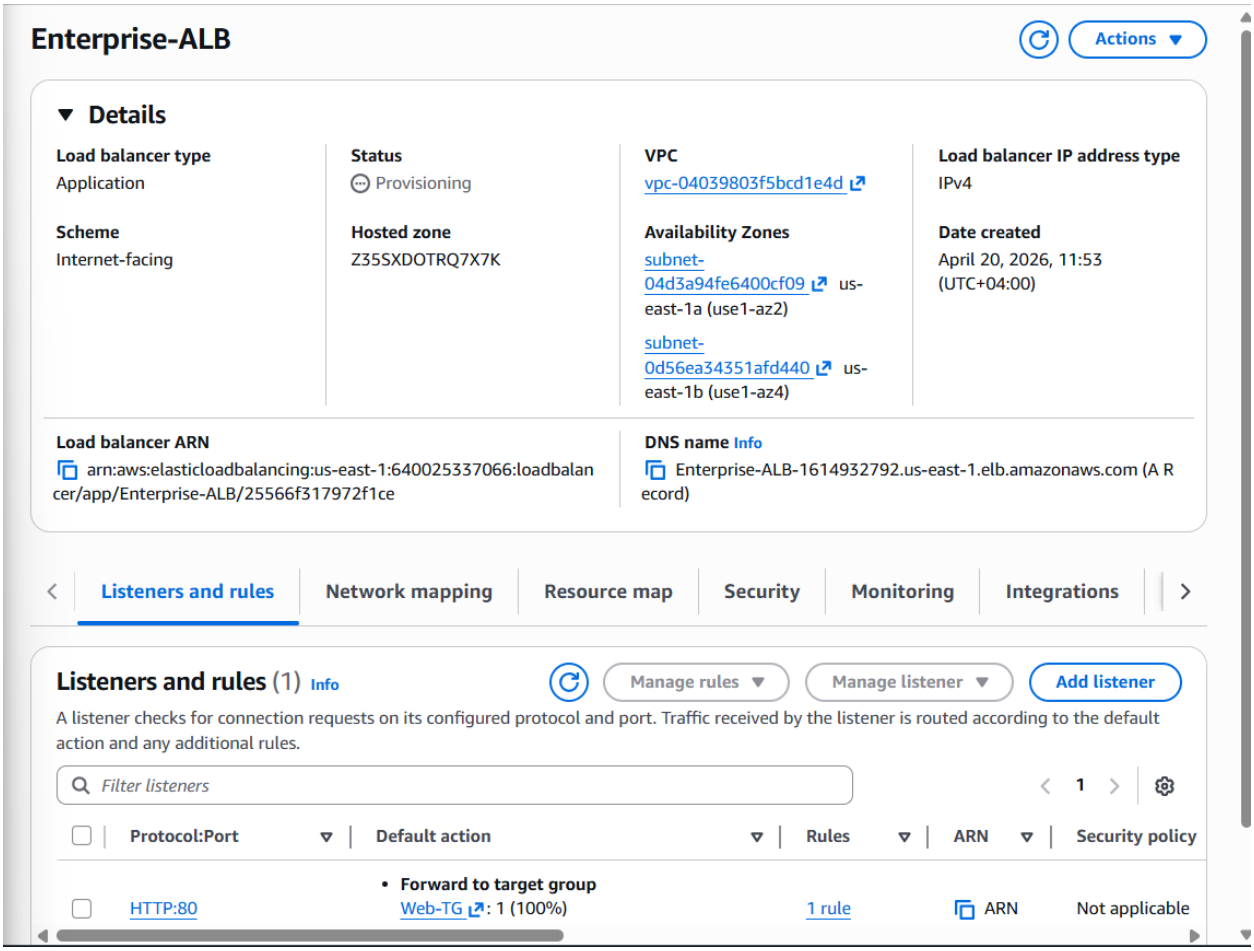
Find security groups by attribute or tag

Name	Security group ID	Security group name
–	sg-061d4d822bd0627a0	Web-Server-SG
–	sg-0c1e033ffb4253df5	DB-SG
–	sg-031b85214a5c30493	default
–	sg-0585d462f691147e7	Puplic-traffic-SG
–	sg-0ce8c36b4422ffad6	default

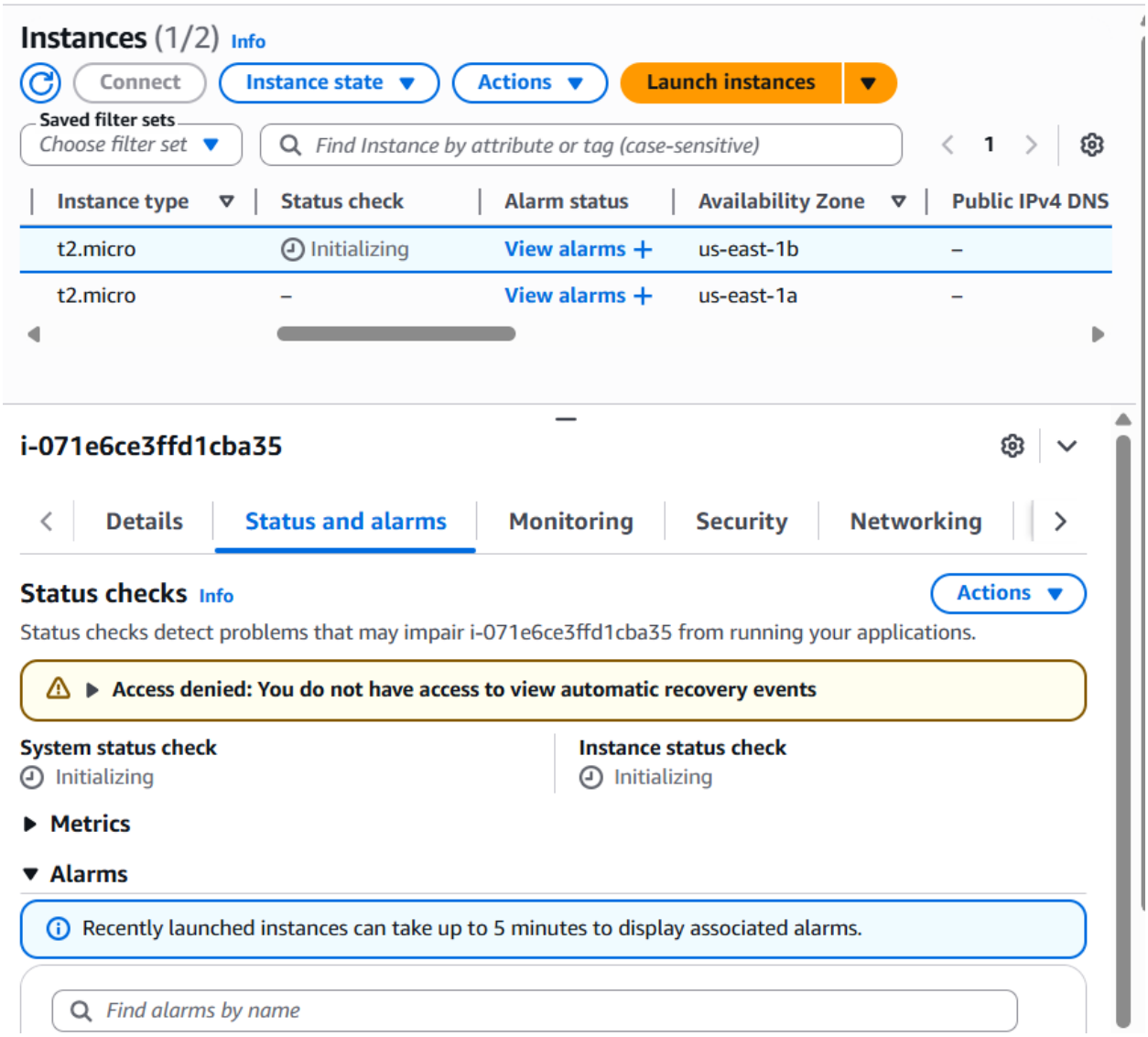
IAM Roles:


EC2-Enterprise-Role Info
Delete
Allows EC2 instances to call AWS services on your behalf.
Summary
Edit
Creation date
April 20, 2026, 11:29 (UTC+04:00)
ARN
arn:aws:iam::640025337066:role/EC2-Enterprise-Role
Last activity
-
Maximum session duration
1 hour
Permissions
Trust relationships
Tags
Last Accessed
Revoke
Permissions policies (1) Info
Simulate
Remove
Add permissions
You can attach up to 10 managed policies.
Filter by Type
Search
All types
1
Policy name
Type
Attached enti...
AmazonSSMMana...
AWS managed
1

Load Balancing



2 EC2 instances



Data Layer

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name

You won't be able to modify the name after your subnet group has been created.

enterprise-db-subnet-group

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

enterprise-db-subnet-group

VPC

Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

Enterprise-VPC (vpc-04039803f5bcd1e4d)
4 Subnets, 2 Availability Zones

Add subnets

Availability Zones

Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

us-east-1a ✕ us-east-1b ✕

Subnets

Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

Instances (1/2) Info

Connect | Instance state | Actions | Launch instances

Saved filter sets
Choose filter set

Find Instance by attribute or tag (case-sensitive)

1

Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
t2.micro	Initializing	View alarms +	us-east-1b	–
t2.micro	–	View alarms +	us-east-1a	–

i-071e6ce3ffd1cba35

Details | Status and alarms | Monitoring | Security | Networking

Status checks Info

Actions

Status checks detect problems that may impair i-071e6ce3ffd1cba35 from running your applications.

Access denied: You do not have access to view automatic recovery events

System status check
Initializing

Instance status check
Initializing

Metrics

Alarms

Recently launched instances can take up to 5 minutes to display associated alarms.

Find alarms by name

Aurora and RDS > Databases > Create database

Create database Info

Engine options

Engine type Info

Choose a database creation method

- ○ Full configuration
  You set all of the configuration options, including ones for availability, security, backups, and maintenance.
- ◉ Easy create
  Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Configuration

DB instance identifier
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

enterprise-db-instance

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Master username Info
Type a login ID for the master user of your DB instance.

admin

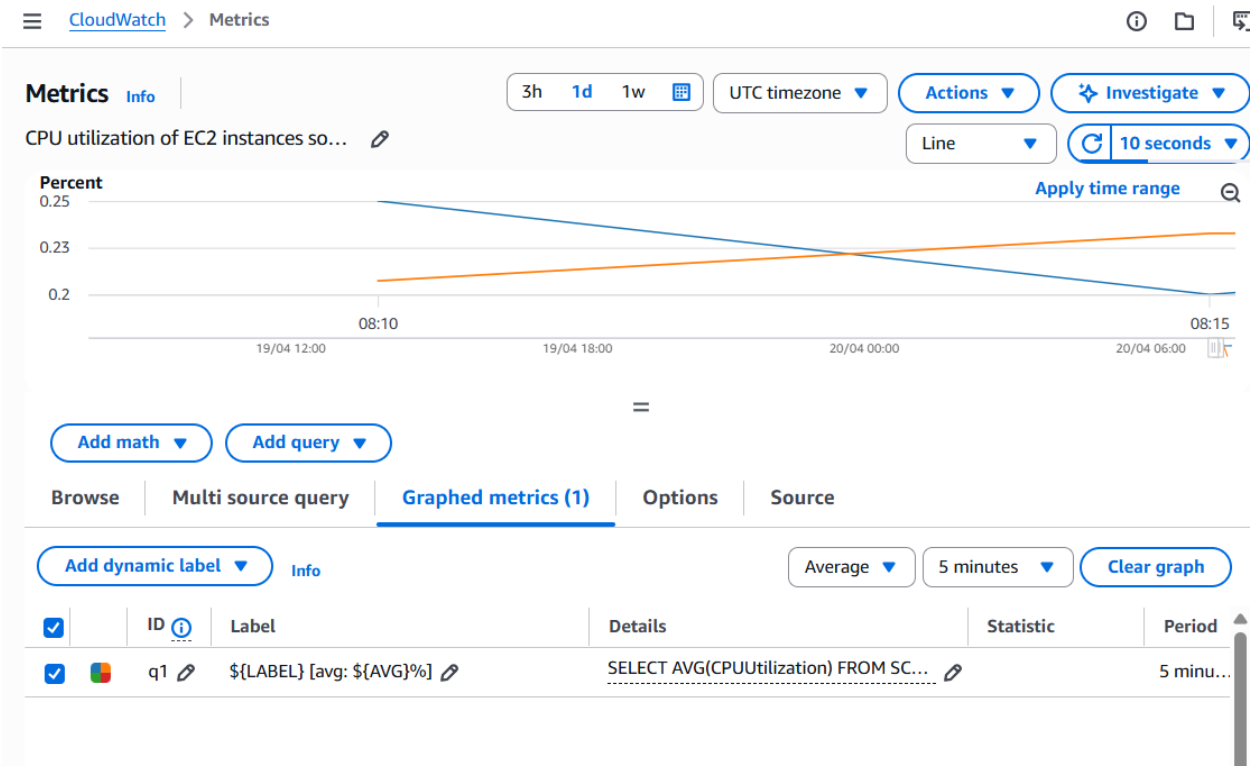
1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

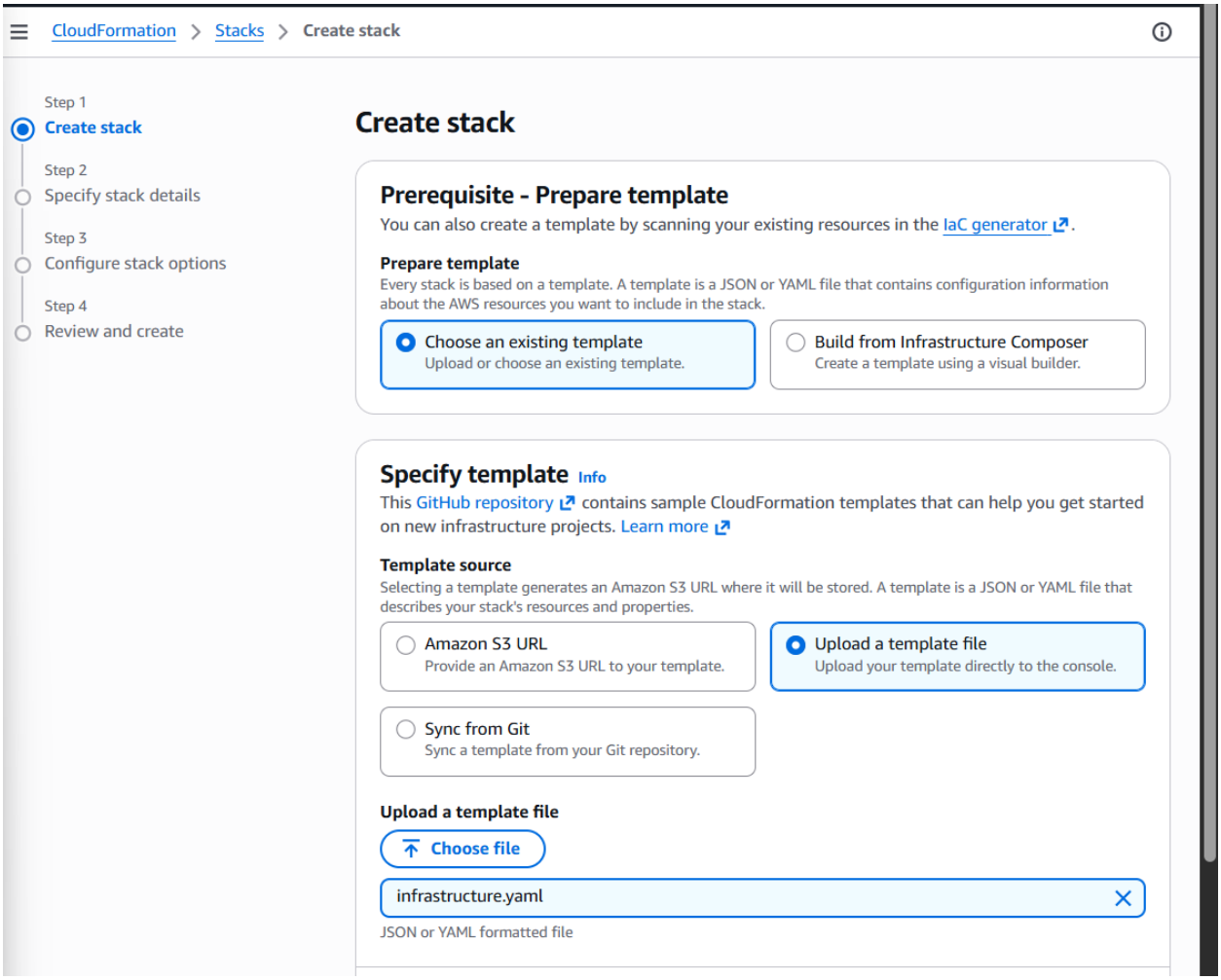
- ◉ Managed in AWS Secrets Manager - *most secure*
  RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
- ○ Self managed
  Create your own password or have RDS create a password that you manage.

Monitoring & Logging

Cloud watch:



Automation


CloudFormation > Stacks > Create stack
Step 1
Create stack
Step 2
Specify stack details
Step 3
Configure stack options
Step 4
Review and create
Create stack
Prerequisite - Prepare template
You can also create a template by scanning your existing resources in the IaC generator.
Prepare template
Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.
Choose an existing template
Upload or choose an existing template.
Build from Infrastructure Composer
Create a template using a visual builder.
Specify template Info
This GitHub repository contains sample CloudFormation templates that can help you get started on new infrastructure projects. Learn more
Template source
Selecting a template generates an Amazon S3 URL where it will be stored. A template is a JSON or YAML file that describes your stack's resources and properties.
Amazon S3 URL
Provide an Amazon S3 URL to your template.
Upload a template file
Upload your template directly to the console.
Sync from Git
Sync a template from your Git repository.
Upload a template file
Choose file
infrastructure.yaml
JSON or YAML formatted file